```python
import string
from pathlib import Path


def clean_words(text):
    text = text.lower()
    text = text.translate(
        str.maketrans("", "", string.punctuation)
    )
    return text.split()

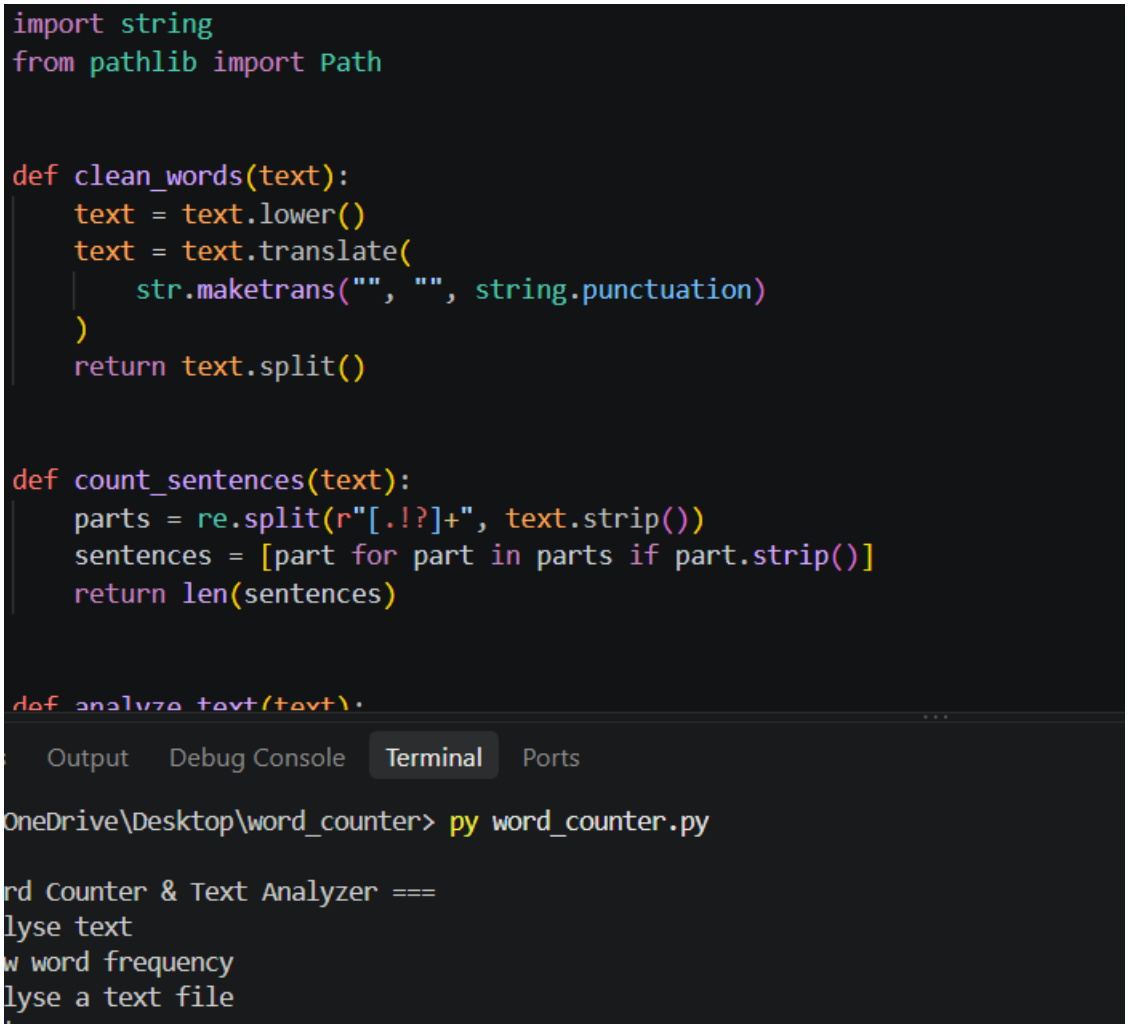

def count_sentences(text):
    parts = re.split(r"[.!?]+", text.strip())
    sentences = [part for part in parts if part.strip()]
    return len(sentences)


def analyze_text(text):
```

Output Debug Console Terminal Ports

```
OneDrive\Desktop\word_counter> py word_counter.py

rd Counter & Text Analyzer ===
lyse text
w word frequency
lyse a text file
```



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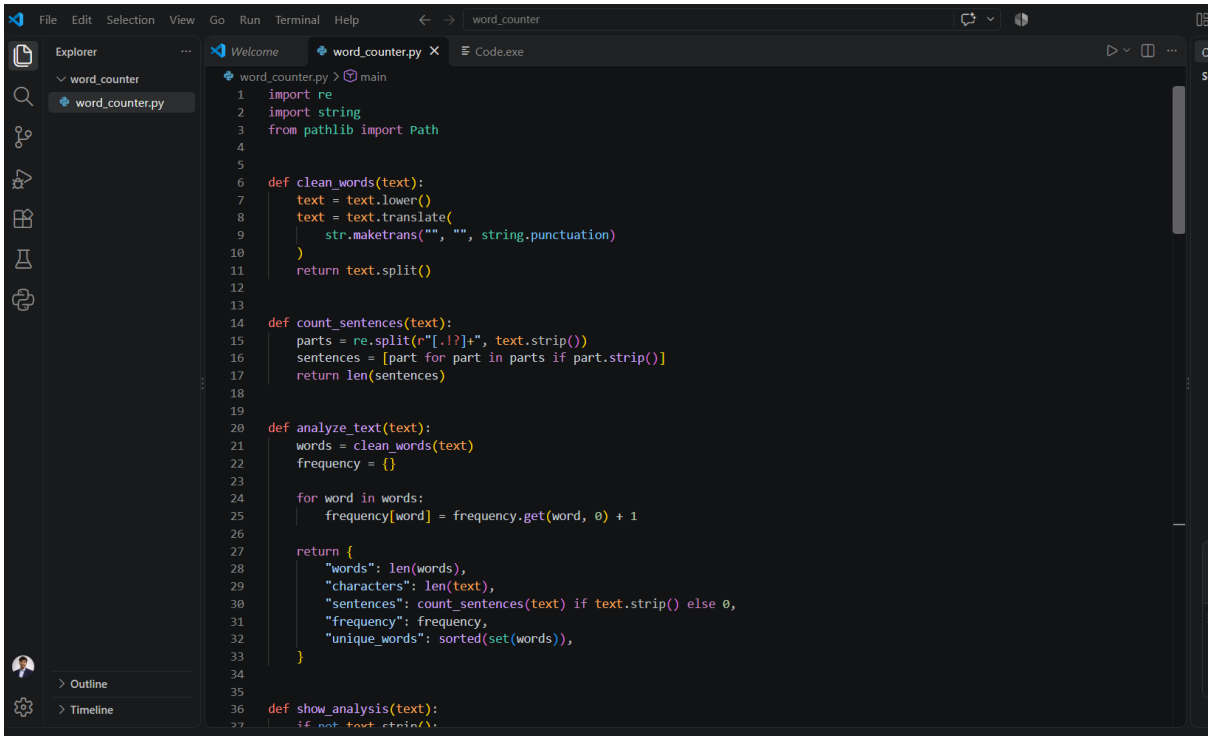

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s  Output  Debug Console  Terminal  Ports

\OneDrive\Desktop\word_counter> py word_counter.py

ord Counter & Text Analyzer ===
alyse text
ow word frequency
alyse a text file
it
 menu option (1-4): 1
 text: python is easy. python is fun.
: 6
cters (with spaces): 30
nces: 2
e words: 4
frequency:
```




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import string
from pathlib import Path


def clean_words(text):
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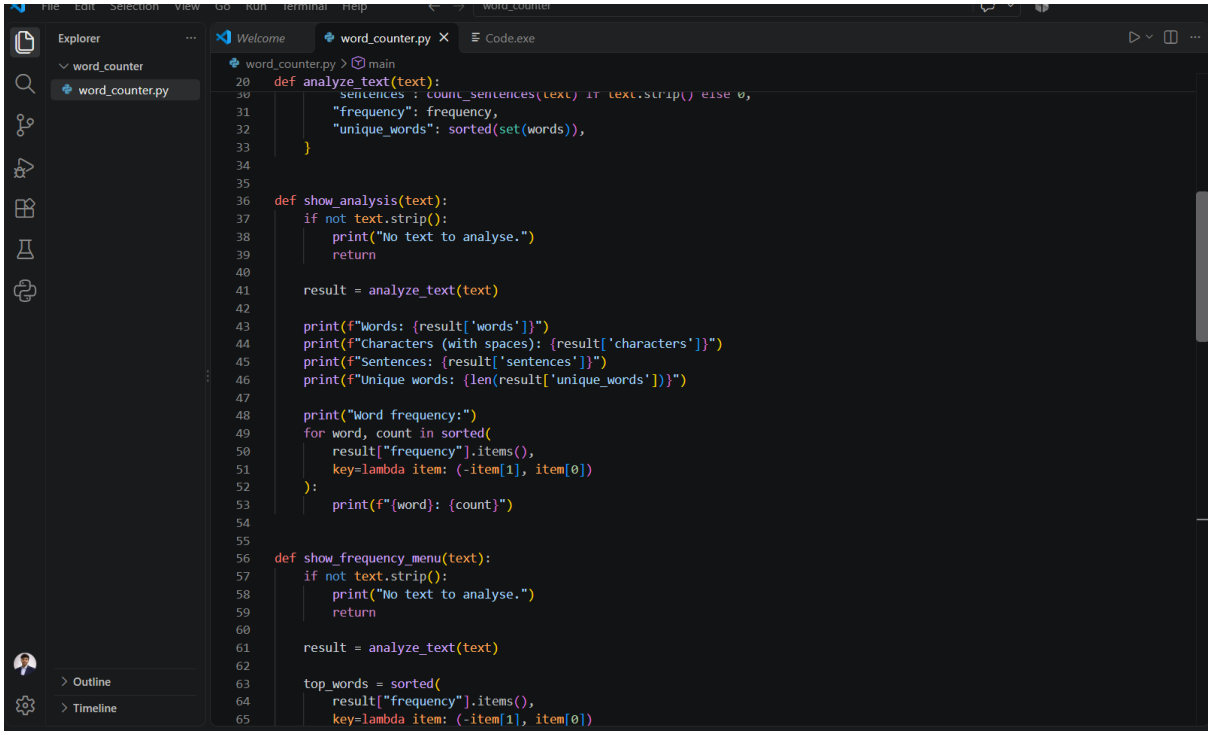

def count_sentences(text):
    parts = re.split(r"[.!?]+", text.strip())
    sentences = [part for part in parts if part.strip()]
    return len(sentences)


def analyze_text(text):
    words = clean_words(text)
    frequency = {}

    for word in words:
        frequency[word] = frequency.get(word, 0) + 1

    return {
        "words": len(words),
        "characters": len(text),
        "sentences": count_sentences(text) if text.strip() else 0,
        "frequency": frequency,
        "unique_words": sorted(set(words)),
    }


def show_analysis(text):
```



```
def analyze_text(text):
        "sentences": count_sentences(text) if text.strip() else 0,
        "frequency": frequency,
        "unique_words": sorted(set(words)),
    }


def show_analysis(text):
    if not text.strip():
        print("No text to analyse.")
        return

    result = analyze_text(text)

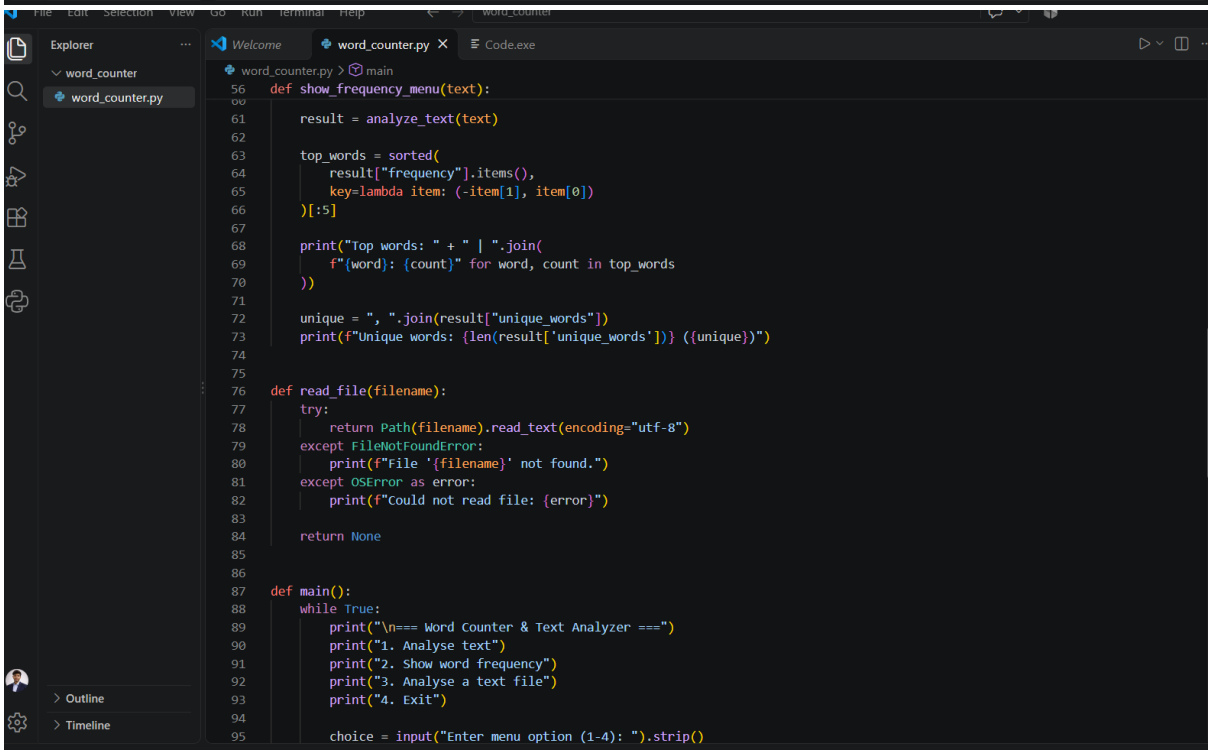
    print(f"Words: {result['words']}")
    print(f"Characters (with spaces): {result['characters']}")
    print(f"Sentences: {result['sentences']}")
    print(f"Unique words: {len(result['unique_words'])}")

    print("Word frequency:")
    for word, count in sorted(
        result["frequency"].items(),
        key=lambda item: (-item[1], item[0])
    ):
        print(f"{word}: {count}")


def show_frequency_menu(text):
    if not text.strip():
        print("No text to analyse.")
        return

    result = analyze_text(text)

    top_words = sorted(
        result["frequency"].items(),
        key=lambda item: (-item[1], item[0])
```



```
def show_frequency_menu(text):
    result = analyze_text(text)

    top_words = sorted(
        result["frequency"].items(),
        key=lambda item: (-item[1], item[0])
    )[:5]

    print("Top words: " + " | ".join(
        f"{word}: {count}" for word, count in top_words
    ))

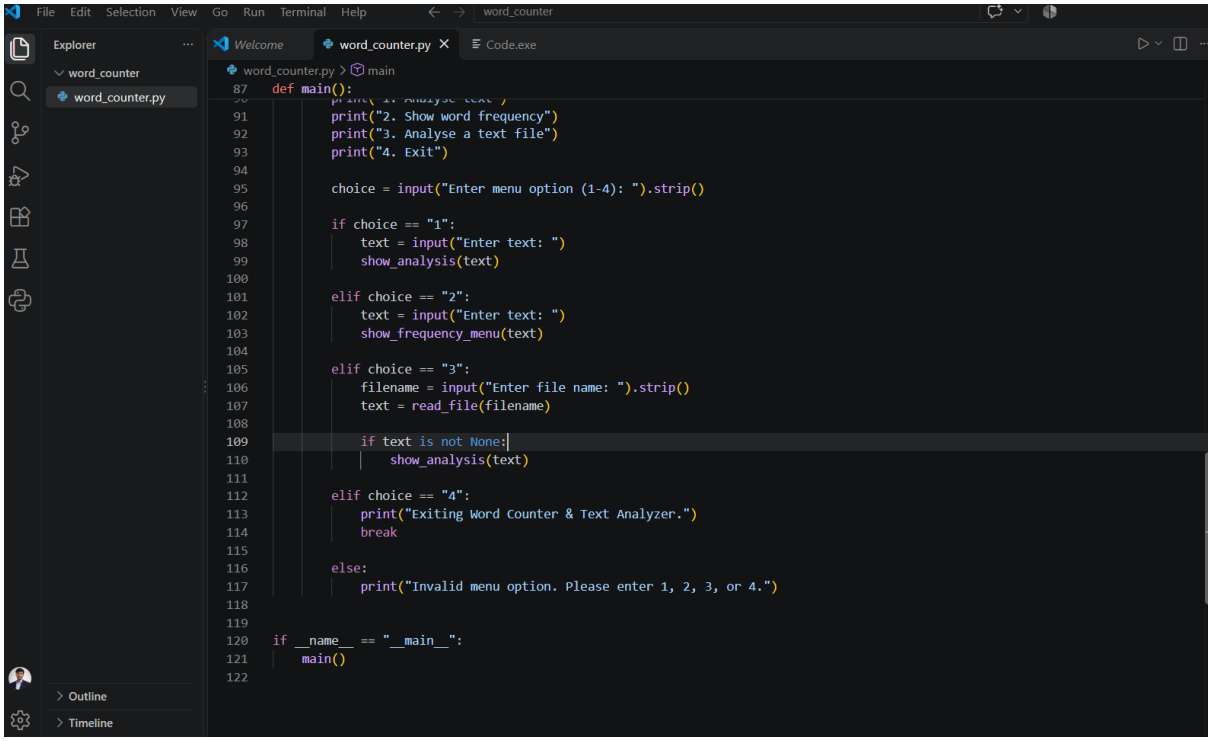
    unique = ", ".join(result["unique_words"])
    print(f"Unique words: {len(result['unique_words'])} ({unique})")

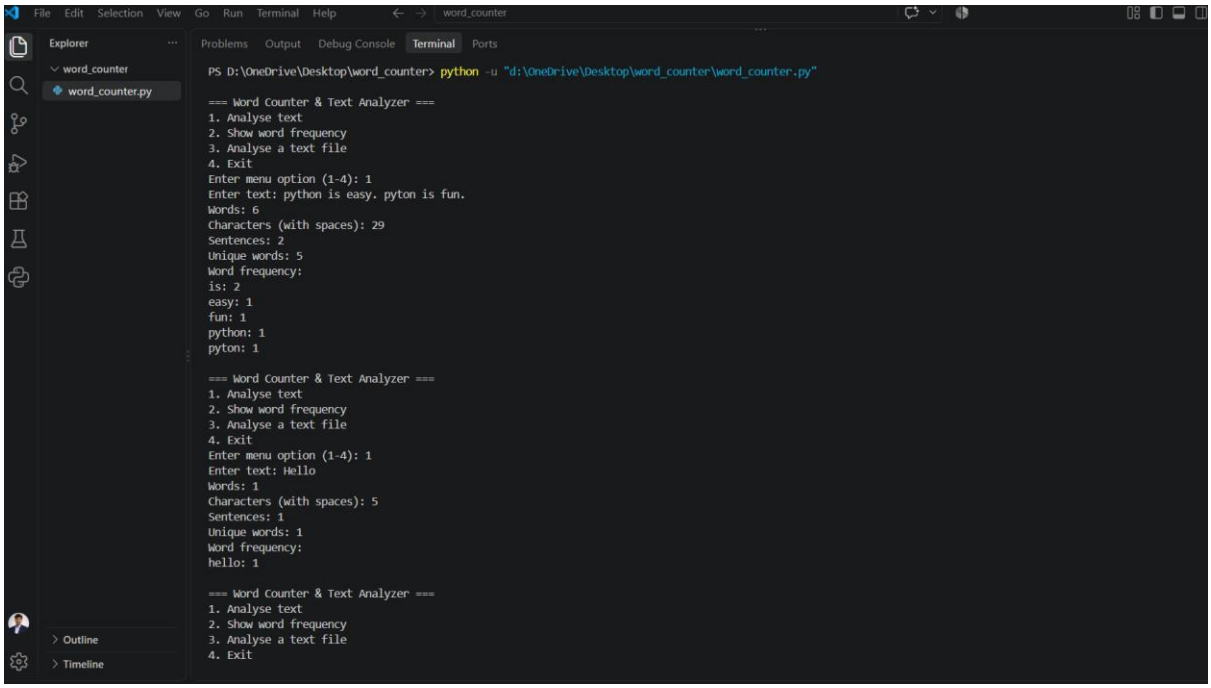

def read_file(filename):
    try:
        return Path(filename).read_text(encoding="utf-8")
    except FileNotFoundError:
        print(f"File '{filename}' not found.")
    except OSError as error:
        print(f"Could not read file: {error}")

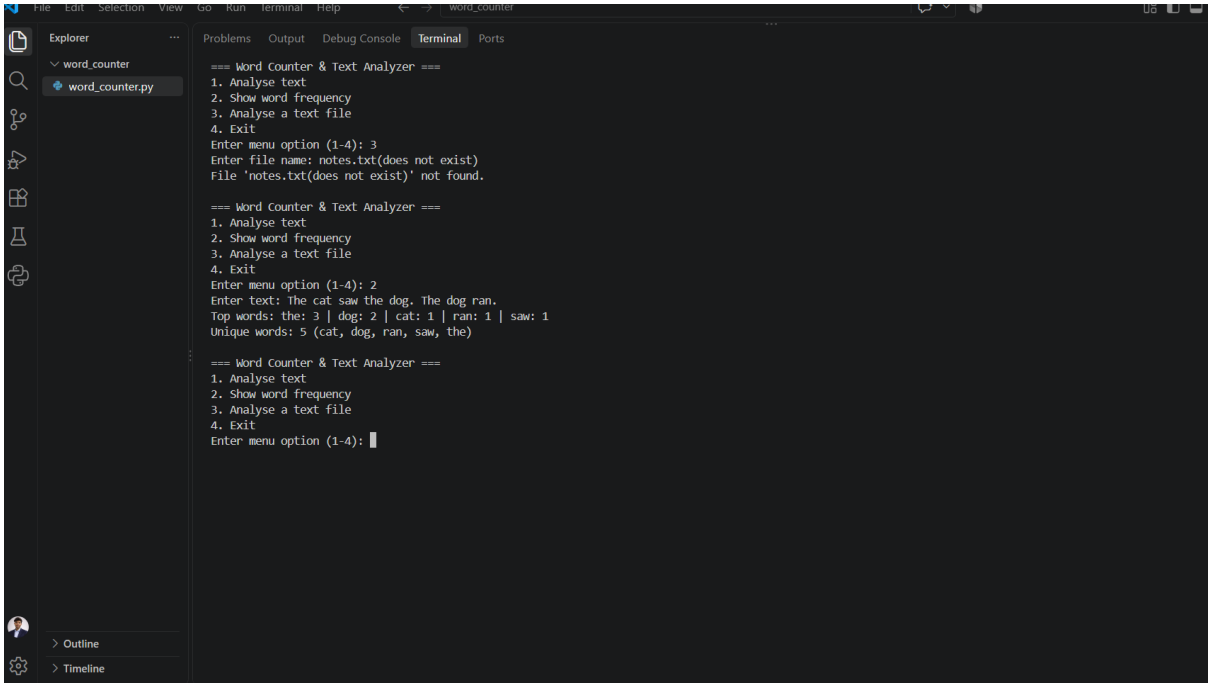
    return None


def main():
    while True:
        print("\n=== Word Counter & Text Analyzer ===")
        print("1. Analyse text")
        print("2. Show word frequency")
        print("3. Analyse a text file")
        print("4. Exit")

        choice = input("Enter menu option (1-4): ").strip()
```


File Edit Selection View Go Run Terminal Help
word_counter
Explorer
word_counter
word_counter.py
Welcome
word_counter.py
Code.exe
word_counter.py > main
def main():
print("2. Show word frequency")
print("3. Analyse a text file")
print("4. Exit")
choice = input("Enter menu option (1-4): ").strip()
if choice == "1":
text = input("Enter text: ")
show_analysis(text)
elif choice == "2":
text = input("Enter text: ")
show_frequency_menu(text)
elif choice == "3":
filename = input("Enter file name: ").strip()
text = read_file(filename)
if text is not None:
show_analysis(text)
elif choice == "4":
print("Exiting Word Counter & Text Analyzer.")
break
else:
print("Invalid menu option. Please enter 1, 2, 3, or 4.")
if __name__ == "__main__":
main()
Outline
Timeline


File Edit Selection View Go Run Terminal Help
word_counter
Explorer
word_counter
word_counter.py
Problems Output Debug Console Terminal Ports
PS D:\OneDrive\Desktop\word_counter> python -u "d:\OneDrive\Desktop\word_counter\word_counter.py"
=== Word Counter & Text Analyzer ===
1. Analyse text
2. Show word frequency
3. Analyse a text file
4. Exit
Enter menu option (1-4): 1
Enter text: python is easy. pyton is fun.
Words: 6
Characters (with spaces): 29
Sentences: 2
Unique words: 5
Word frequency:
is: 2
easy: 1
fun: 1
python: 1
pyton: 1
=== Word Counter & Text Analyzer ===
1. Analyse text
2. Show word frequency
3. Analyse a text file
4. Exit
Enter menu option (1-4): 1
Enter text: Hello
Words: 1
Characters (with spaces): 5
Sentences: 1
Unique words: 1
Word frequency:
hello: 1
=== Word Counter & Text Analyzer ===
1. Analyse text
2. Show word frequency
3. Analyse a text file
4. Exit
Outline
Timeline


File Edit Selection View Go Run Terminal Help
word_counter
Explorer
word_counter
word_counter.py
Problems Output Debug Console Terminal Ports
=== Word Counter & Text Analyzer ===
1. Analyse text
2. Show word frequency
3. Analyse a text file
4. Exit
Enter menu option (1-4): 3
Enter file name: notes.txt(does not exist)
File 'notes.txt(does not exist)' not found.
=== Word Counter & Text Analyzer ===
1. Analyse text
2. Show word frequency
3. Analyse a text file
4. Exit
Enter menu option (1-4): 2
Enter text: The cat saw the dog. The dog ran.
Top words: the: 3 | dog: 2 | cat: 1 | ran: 1 | saw: 1
Unique words: 5 (cat, dog, ran, saw, the)
=== Word Counter & Text Analyzer ===
1. Analyse text
2. Show word frequency
3. Analyse a text file
4. Exit
Enter menu option (1-4):
Outline
Timeline